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Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 33706 Roll No. | | | | | | | | | | |

B. Tech. Examination 2021-22

(Even Semester)

DATA STRUCTURE USING 'C'

Time : Three Hours] *[Maximum Marks : 60*

Note :– Attempt all questions.

SECTION – A

1. Attempt all parts of the following : 8×1=8

   (a) What is data structure and why we use it?

   (b) Define abstract data type.

   (c) Compare time and space complexity.

   (d) If the tower of Hanoi is operated on n = 10 disks, calculate the total number of moves.

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(e) Write pseudo code for merging two single linked list.

(f) Write the use of priortiy queue in computer science.

(g) What is Sorting? How is sorting essential for data base application?

(h) When does a graph become tree?

SECTION – B

2. Attempt any two parts of the following : $2 \times 6 = 12$

(a) What are the various asymptotic notations? Explain the Big Oh notation.

(b) Convert it into prefix and postfix expressions :

$$\big((a+b)+c*(d+e)+f\big)*(g+h)$$

(c) Differentiate between internal and external sorting. Also differentiate between linear search and bianry search.

(d) What are the different representations of graph in data structure? Explain.

SECTION – C

Note :– Attempt all questions. Attempt any two parts from each questions. 5×8 = 40

3. (a) What is linked list? Explain different types of linked list. How to insert a node in the middle of linked list of size-10?

   (b) Write a recursive "C" functions to implement binary search and compare its time complexity.

   (c) Write the algorithm for merge sort.

4. (a) Explain and classify the data structure. Write the different types of operation used in data structure.

   (b) Differentiate between recursion and iteration.

   (c) A very large array of elements is to be sorted. The program is to be run on a personal computer with limited memory. Which sort would be a better choice : Heap sort or quick short? Why?

5. (a) Explain different types of traversing in tree. Find the preorder traversing for given post order and in order traversing :

In Order : D G B A H E I C F

Post Order : G D B H I E F C A

(b) What is Quick Sort? Soft the given values using quick sort. Show all the steps / iterations :

38, 81, 22, 48, 13, 69, 93, 14, 45, 58, 79, 72

(c) What is Stack? Write the algorithm for PUSH and POP operation.

6. (a) Explain with diagram the functioning of doubly linked list.

(b) Explain all the collision resolution strategies in hashing.

(c) Discuss Prim's and Kruskal's algorithm with suitable example.

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